

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and executed electrical power system studies, including **load flow**, **short circuit**, and **protection coordination** analysis for critical infrastructure projects, ensuring adherence to client specifications.
- Developed and validated **power architectures** for large-scale data centre facilities, encompassing utility intake, MV/LV distribution, **UPS topologies**, and generator integration.
- Managed the complete electrical design lifecycle from feasibility and concept through to detailed design, procurement support, construction, commissioning, and handover for complex technology projects.
- Integrated **monitoring and controls** systems (BMS/EPMS/SCADA) and defined metering strategies to align with high-availability objectives, such as Tier III and Tier IV standards.
- Assessed and embedded **decarbonization and energy efficiency** strategies, including high-efficiency UPS selection and advanced control systems for data centre operations.
- Coordinated multi-disciplinary engineering interfaces, including mechanical, structural, and ICT, to ensure the delivery of fully integrated and resilient designs.
- Mentored and guided a team of junior electrical engineers on technical best practices, quality standards, and project delivery methodologies within a fast-track environment.

SKILLS

Electrical Design – Power Architectures, Utility Intake, MV/LV Distribution, UPS Topologies, Generators, Switchgear, PDUs, STS, Busway, Earthing, Lighting, Power Systems Studies, Protection Design, Load Flow, Short Circuit, Protection Coordination, Arc-Flash, Harmonic Analysis

Data Centre Technologies – Hyperscale, Mission Critical Environments, Resilience Features, Availability Objectives, Decarbonization, Energy Efficiency, Grid-Interactive Strategies

Standards & Compliance – IEC/BS/EN, NEC/NFPA, Client Specifications, AECOM Quality Procedures

Tools & Software – ETAP, SKM, BMS, EPMS, SCADA, Jenkins, Git, Jira, Confluence

Project Management – Feasibility, Concept Design, Detailed Design, Procurement Support, Construction, Commissioning, Handover, Fast-Track, Design-Build, Team Leadership, Commercial Acumen, Project Financials

Programming & Scripting – Python, Java, SQL, Spring Boot, REST APIs, AWS, Docker, Linux

PROJECTS

Data Centre Electrical Infrastructure Design

- Engineered the complete electrical design for a new hyperscale data centre, covering utility substations, **MV distribution**, and **UPS systems** to meet stringent uptime requirements.
- Performed detailed **power systems analysis**, including **arc-flash studies** and **harmonic analysis**, using ETAP to ensure safety and compliance with NEC standards.
- Collaborated with mechanical and ICT teams to integrate power delivery with cooling and networking infrastructure, ensuring seamless operation across all systems.

Energy Efficiency Program for Enterprise Data Facilities

- Developed and implemented a strategy to enhance energy efficiency in existing data centres by integrating advanced **control systems** and optimizing **UPS topologies**.
- Analyzed existing power consumption data and proposed upgrades to **high-efficiency UPS** and **demand response** mechanisms, projecting a **15% reduction** in energy usage.
- Authored technical documentation and presented findings to client stakeholders, outlining the benefits of adopting **grid-interactive strategies** for operational cost savings and sustainability.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)